Meteograms

The model weather of one place over the days — on the Weather page and on the phone

Otso Laakso (OH2GAX)
Finland · near Helsinki-Vantaa airport (EFHK)

In short. A **meteogram** shows the weather of *one place* hour by hour on one time axis: the temperature, the precipitation, the wind, the pressure and the clouds, one panel under the other, from the past hours to up to 16 days ahead. MODE-S Wind draws it from two weather models — **MEPS** (MET Norway, the next two and a half days in fine detail) and **GFS** (NOAA, the whole world, 16 days) — and puts next to them what was actually **measured**: the aircraft's own temperatures, winds and freezing level over southern Finland, the landings' winds and the airport's METAR.

- On the **Weather page**: Map | Meteogram at the start of the control strip (Section 2).
- On the **phone** (`/app`): the Meteogram view, the same drawing made for a narrow screen (Section 8).
- Every choice — the place, the model, the range, the panels and the options — is remembered in the browser.
- How far the forecast can be trusted is shown too: the **typical error** band and table, from the forecasts checked against the airport's METAR (Section 7).

Section 4 goes through the chart panel by panel, Section 5 explains the measured values and Section 10 how to read a meteogram sensibly.

For viewing only. A meteogram is a model's weather at the model's grid cell of the place, not an observation or an official forecast. For flight operations use the official products (TAF, GAFOR, the aviation weather briefing). **All figures show real data**, drawn in a test with the same page code: the live server's model runs of 7 November 2026 (GFS 00 UTC, MEPS 03–09 UTC) and its forecast check of the last 30 days as they were at 11:32 UTC, and the stored data of 6 November 2026 with the real MEPS runs of 03–12 UTC and the GFS run of 12 UTC of that day (drawn at 16:46 UTC). The newest model values (dew point, visibility, freezing level, CAPE, the 850 hPa wind) were stored only from the evening of 7 November, so the figures of 7 November do not have them yet.

Contents

1 What a meteogram shows **2**

2 Opening a meteogram **2**

3 The two models **3**

4 Reading the chart, panel by panel **4**

5 What MODE-S Wind measures itself **7**

6 The values of one hour **8**

7 How far can the forecast be trusted? **8**

8 On the phone **9**

9 In Finnish **10**

10 Reading a meteogram sensibly **10**

A Terms in English and Finnish **11**

1 What a meteogram shows

A weather map shows one moment everywhere; a meteogram shows **one place at every moment**. Time runs from left to right in **UTC** (Finnish winter time is UTC+2, summer time UTC+3), with the days named at the top and the hours under them. Every panel uses the same time axis, so a front shows as the temperature falling, the wind veering and strengthening, the pressure reaching its lowest and the rain starting — all at the same place on the axis (Fig. 1).

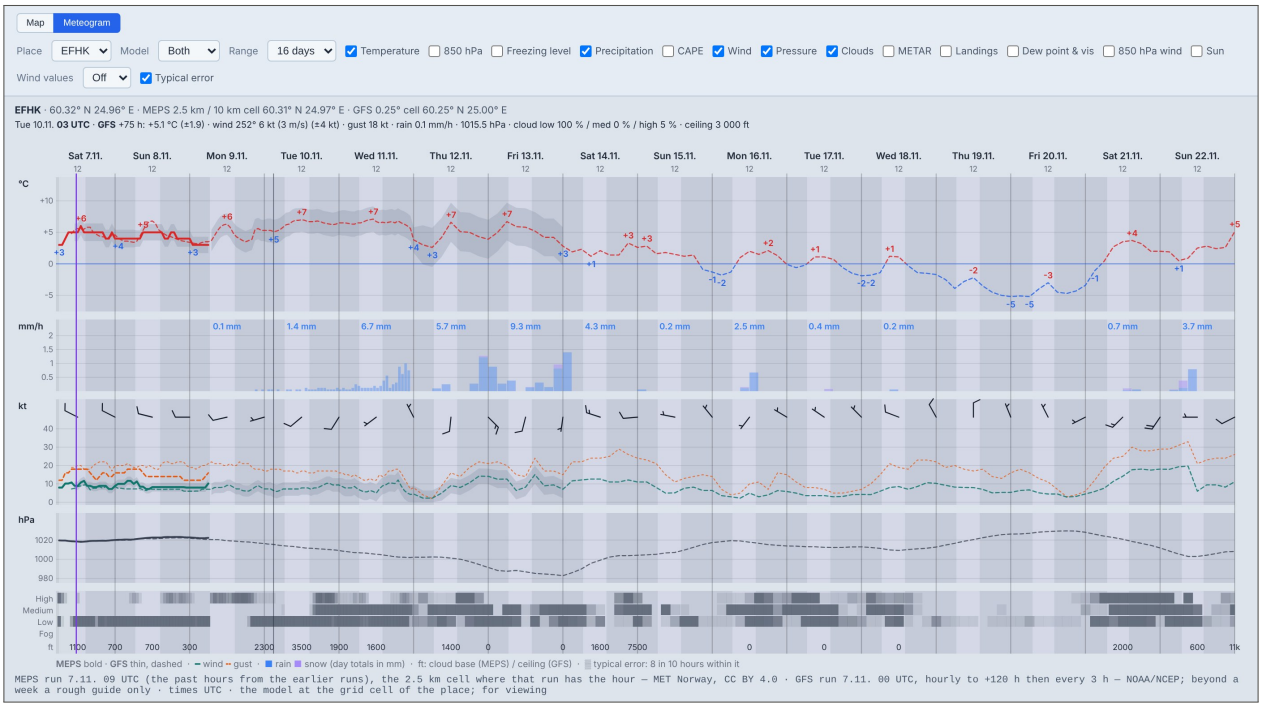


Figure 1. The Weather page's meteogram at EFHK: Model Both, Range 16 days, the default panels and the Typical error band; the live server's runs of 7 November 2026 (GFS 00 UTC, MEPS 09 UTC with the past hours from the earlier runs), drawn at 11:32 UTC, light theme. The mouse is on Tuesday 10 November 03 UTC: the line above the chart gives that hour's values. MEPS (bold) covers the first two days, GFS (thin, dashed) the whole range; the cold spell around 18–20 November is GFS's view twelve days ahead — a rough guide only (Section 10).

Common to all panels:

- **Night** is shaded — computed from the sun's elevation at the place, so the shading follows the place's real day length.
- The **forecast** part (after now) is lightly tinted; the **now line** is the violet vertical line.
- **Day lines** separate the days (midnight UTC).
- With Both models, **MEPS is bold** and **GFS thin and dashed**. Where MEPS has the hour, the barbs, the cloud rows and the day totals come from MEPS, after that from GFS.
- The **legend** under the chart names every line and symbol that is drawn; it changes with the panels and options in use.
- The line **above the chart** gives the place, its position and the model grid cells; the line **under it** the model runs used, the data sources and the time zone.

2 Opening a meteogram

2.1 On the Weather page

Map | Meteogram at the left end of the Weather page's control strip switches between the weather map and the meteogram. The meteogram's own controls then take the strip (Table 1); a ? at their end opens this guide.

The place (Place):

- **EFHK** — the airport (the default). Only here, and within 20 km of it, are the METAR and the landings' winds shown, and the typical error is measured here.
- **Receiver sites** — the MODE-S Wind receivers (RX-01, RX-02 . . .), the same as the triangles on the maps, at their position rounded to 0.01° (about 1 km).
- **A pinned place** — any place on the map: switch on Point values on the map, click the place and press Meteogram in its card. The place stays in the list (*60.45° N 24.80° E (pinned)*) until you pin another.

An address for a place. The browser's address follows the meteogram: `/weather#mg` opens the airport's,

`/weather#mg=60.450,24.800` a place by its latitude and longitude and `/weather#mg=RX-02` a receiver site — handy as a bookmark or to send to someone with access.

The meteogram is refreshed every 10 minutes while it is shown; a new model run comes in on its own (MEPS every 3 hours, GFS every 6 hours, a few hours after the run's start time).

2.2 On the phone

On the phone view (`/app`) the Meteogram tile opens the same drawing for a narrow screen, with the place, model and range as chips and the other choices in the menu (Section 8). A place tapped on the phone's Weather map opens with Meteogram here › in its card and then appears as Tapped.

2.3 The controls

Control	Choices	What it does
Place	**EFHK** / receiver sites / pinned place	The place of the meteogram (above)
Model	MEPS / GFS / **Both**	Which model(s); GFS only on servers that fetch it (Section 3)
Range	MEPS: past 24 h / 3 / 7 days + forecast	With MEPS: how far back; the forecast is MEPS's own (about 2.5 days)
	GFS, Both: 2 / 5 / 7 / 10 / **16 days**	How far ahead
Panels	**Temperature**, **Precipitation**, **Wind**, **Pressure**, **Clouds** (on); Freezing level, CAPE (off)	Each panel on or off (Section 4)
Options	850 hPa, METAR, Landings, Dew point & vis, 850 hPa wind, Sun (off)	More in the panels (Sections 4 and 5)
Wind values	**Off** / kt / m/s	The wind direction and speed in numbers under the barbs
Typical error	**on** / off	The band of how far off the forecast has been, and the table under the chart (Section 7)

Table 1. The meteogram's controls on the Weather page (the defaults in bold). Hovering a control shows a short explanation.

3 The two models

	MEPS	**GFS**
Made by	MET Norway (MetCoOp, the joint model of the Nordic and Baltic weather services)	NOAA / NCEP (USA)
Area	the Nordic countries	the whole world
Grid	2.5 km (near the airport) and 10 km	0.25° (about 28 km north–south, 14 km east–west here)
Runs	every 3 hours; the main runs (00, 06, 12, 18 UTC) about 2.5 days ahead	every 6 hours (00, 06, 12, 18 UTC), 16 days ahead
Steps	1 hour	1 hour to +120 h, then 3 hours
Past hours	yes: the first hours of the earlier runs	no (from the hour before now)
Best for	the next two days: showers, the wind near the coast, low cloud	the trend over the week; beyond a week a rough guide only
Licence	CC BY 4.0	public domain

Table 2. MEPS and GFS in the meteogram.

Both draws GFS over its whole range (thin, dashed) and MEPS over its own hours (bold). Where the two agree, the forecast is more likely to hold; where they part, the weather is less certain. **MEPS alone** also shows the *past* days: each past hour comes from the newest run that had it (so the past part is close to an analysis of what happened, not an old forecast) — see Fig. 2.

The grid cell. The values are those of the model's grid cell that contains the place: MEPS's 2.5 km cell near the airport (where the newest run has the hour), otherwise its 10 km cell; GFS's 0.25° cell. The line above the chart gives the cell centres (*MEPS 2.5 km / 10 km cell 60.31° N 24.97° E · GFS 0.25° cell 60.25° N 25.00° E*). The 850 hPa values, the dew point, the visibility, the freezing level and CAPE always come from MEPS's 10 km grid.



Figure 2. MEPS alone, past 24 h + forecast, with 850 hPa and METAR on, dark theme; the stored data of 6 November 2026 and the real MEPS run of 12 UTC, drawn at 16:46 UTC, the mouse on Saturday 14 UTC. The past day on the left: MEPS's past hours with the METAR's temperatures (dots), winds and QNH, and the 850 hPa temperature measured by the aircraft (fuchsia dots) next to MEPS's (fuchsia line).

4 Reading the chart, panel by panel

Fig. 3 shows every panel and option at once; Figs. 4 and 5 enlarge the past part of it.

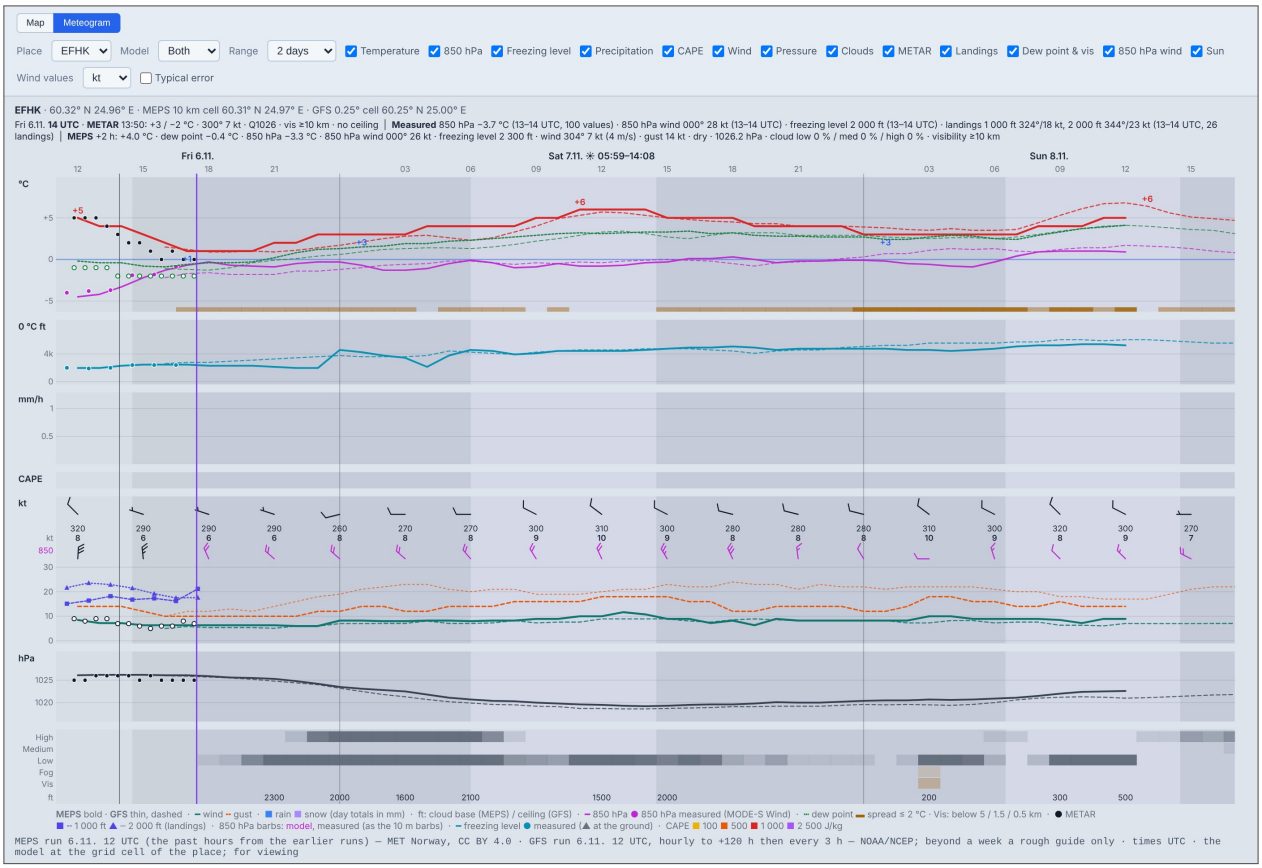


Figure 3. Every panel and option on: Both, 2 days, Wind values kt, light theme; the stored data of 6 November 2026 and the real MEPS and GFS runs of that day (MEPS 12 UTC, GFS 12 UTC), drawn at 16:46 UTC. The mouse is on Friday 14 UTC, in the past: the read-out gives the METAR, what the aircraft measured and MEPS's values of that hour. *Sat 7.11. … 05:59–14:08*: the sunrise and sunset.

4.1 Temperature

The 2 m temperature in °C, **red above and blue below 0 °C**, with each day's highest and lowest written at the curve. The 0 °C line is drawn when the scale crosses it. Options in this panel:

- 850 hPa — the temperature at 850 hPa, about 1 500 m or 4 800 ft up, on the same scale in **fuchsia**: the models' line and, for the past hours, the value **measured by the aircraft** (fuchsia dots; Section 5). 850 hPa is the usual level for telling the air mass: with −5 °C or colder there, falling precipitation usually reaches the ground as snow; a rising 850 hPa temperature shows warm air coming in before it reaches the ground.
- Dew point & vis — the dew point (green, dotted) and, at the foot of the panel, a **strip where the temperature is within 2 °C of the dew point** (darker within 1 °C): the air is nearly saturated, mist or fog is possible. The same option adds a *Vis* row under the clouds (below).
- METAR — the airport's observed temperature (dots) and dew point (open green circles) in the past hours.

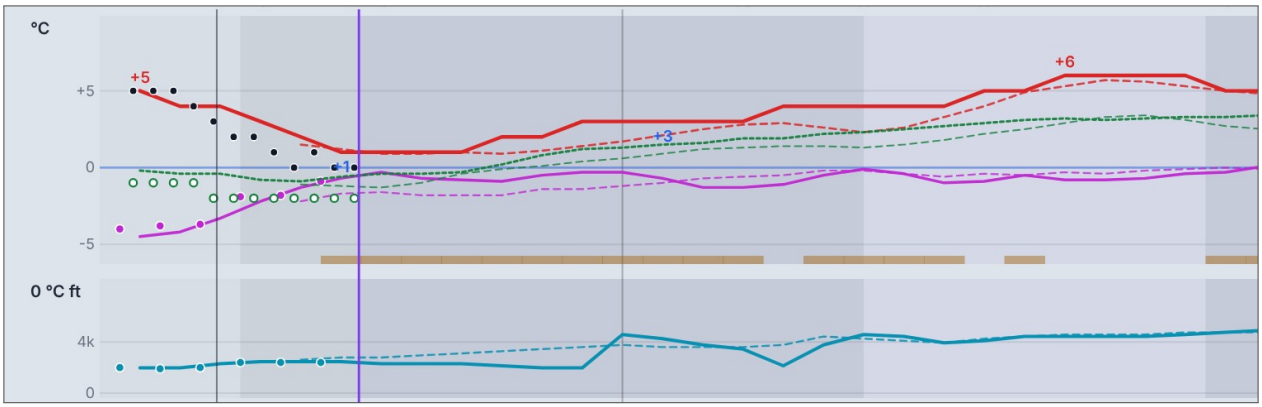


Figure 4. Fig. 3 enlarged: the temperature and freezing-level panels on 6 November. The METAR temperatures (black dots) fall from +5 to 0 °C in the afternoon, with MEPS (bold red) and GFS (dashed) a little warmer; the dew points (open green) are −1 to −2 °C. Fuchsia: the 850 hPa temperature — the dots measured by the aircraft (−4 to −1 °C), the lines from MEPS and GFS. The brown strip ahead: the dew point within 2 °C of the temperature. Below: the freezing level, measured about 2 000 ft (cyan dots), MEPS rising to about 4 000 ft overnight.

4.2 Freezing level

A panel of its own (off by default): the height of the 0 °C level in feet above sea level — the models' line (cyan) and, for the past hours, the level the aircraft **measured** (dots; a triangle at the ground when the whole column was below 0 °C). It matters for icing in cloud, and for whether precipitation falls as rain or snow: with the freezing level below about 1 000 ft, snow usually reaches the ground. When a warm layer lies above a cold one, the level shown is the highest one (the model's *from above*).

4.3 Precipitation

The rain (blue) and snow (violet) of each hour as bars in **mm per hour**, with the **day's total** in mm at the top of each day. After +120 h GFS gives 3-hour steps, drawn as wider bars of the step's mean. With Both, MEPS's bars are narrow in front of GFS's.

4.4 Thunderstorm (CAPE)

A thin strip (off by default) coloured by the model's **CAPE**, the convective available potential energy — the energy a rising parcel of air would gain: yellow from 100, orange from 500, red from 1 000 and violet from 2 500 J/kg. CAPE says that showers and thunderstorms *can* grow, not that they will (something must lift the air). Mostly empty in winter.

4.5 Wind

The 10 m wind (teal) and the gust (orange, dashed) in knots, with **wind barbs** along the top. The staff of a barb points to where the wind blows *from* (a staff pointing up: wind from the north); each short line is 5 kt, each long line 10 kt and a pennant 50 kt; a small circle is calm (under 2.5 kt). 1 kt = 0.514 m/s, so 10 kt is about 5 m/s. More in the wind panel:

- Wind values kt / m/s — the wind **in numbers** under each barb: the direction rounded to 10° (*360* for north) and, under it, the speed in knots or metres per second; the unit is at the left end of the row. The same model hour as the barb above it; calm shows as – over *0*.
- 850 hPa wind — a second row of barbs at about 1 500 m: the model's (fuchsia) and, in the past hours, the one the aircraft measured (in the colour of the 10 m barbs).
- Landings — the wind the landing aircraft met at **1 000 ft** (squares) and **2 000 ft** (triangles) above their runway: the median of each hour's landings at EFHK. Often much stronger than the 10 m wind — a measure of the low-level wind shear on the approach.
- METAR — the observed mean wind (open circles) and gust (orange dots).

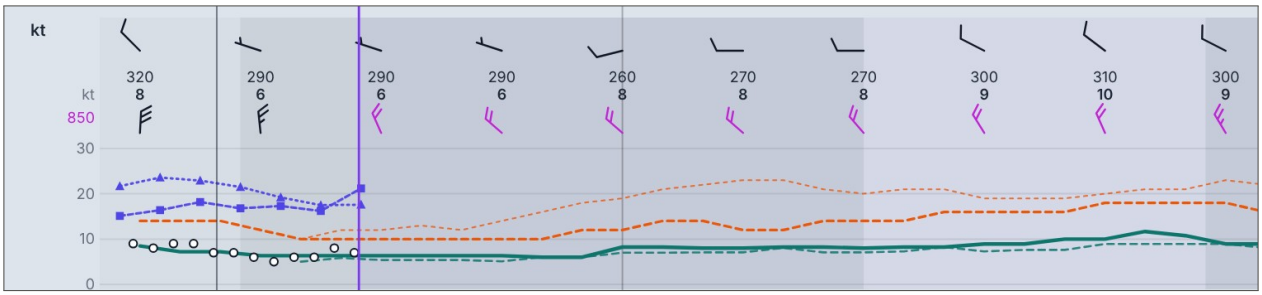


Figure 5. Fig. 3 enlarged: the wind panel on 6 November with Wind values kt. Under the 10 m barbs the direction and speed (*320 / 8 …*), under them the 850 hPa barbs — measured from the aircraft in the past hours (black), MEPS and GFS after now (fuchsia), from the north at about 25–30 kt. In the past part the landings' winds at 1 000 ft (squares, about 15–18 kt) and 2 000 ft (triangles, about 20–24 kt) are two to three times the METAR's 10 m wind (open circles, 6–9 kt).

4.6 Pressure

The sea-level pressure in hPa (with METAR: the QNH as dots). A falling pressure brings a low or a front nearer; the pressure is the model value that has been the most reliable in the forecast check (Section 7).

4.7 Clouds

Rows of the **high**, **medium** and **low** cloud cover (and **fog** with MEPS), darker the more cover. With Dew point & vis a *Vis* row shows where the visibility is below 5 km (darker below 1.5 km and 500 m). The numbers under the rows (*ft*) give in feet:

- from **MEPS** the **cloud base** — the lowest cloud of any amount, shown where there is low cloud;
- from **GFS** the **ceiling** — the base of the lowest broken or overcast layer.

Bases above about 20 000 ft are not shown.

4.8 Sun

With Sun each day's label gives the sunrise and sunset at the place in UTC (*Sat 7.11. … 05:59–14:08*), when the days are wide enough — the shorter ranges. In the polar night or the midnight sun it says *none* or *all day*.

5 What MODE-S Wind measures itself

The dots and the extra barbs in the past part are not from a model. They come from the replies of the aircraft flying over southern Finland, decoded by the MODE-S Wind receivers (the temperature and wind each aircraft computes and sends when the ground radars ask), and from the airport's own observations.

Value	Where it comes from	Shown for
850 hPa temperature	the hourly profile of the area: the temperatures the aircraft reported between 4 000 and 6 000 ft pressure altitude, taken to 850 hPa; the read-out gives the number of values	places within 150 km of the receiver
850 hPa wind	the same profile: the 4 000–5 000 and 5 000–6 000 ft layers, interpolated to 850 hPa	same
Freezing level	from the ground up: the airport's METAR, the climb-outs and landings at the airport, then the area's aircraft above; a warm layer aloft counts only from +0.5 °C	same
Landings' winds	the band winds of each landing at EFHK (the approach history), median of the hour at 1 000 and 2 000 ft above the runway	within 20 km of EFHK
METAR	the EFHK METARs and SPECIs (temperature, dew point, wind, gust, QNH, visibility, ceiling)	within 20 km of EFHK

Table 3. The measured values in the meteogram.

Why it is useful. The model's hours of the past are the model's best guess; the measured values show whether it was right. On 6 November 2026, for example, the aircraft measured the freezing level at 1 800–3 100 ft while MEPS had 2 000–2 500 ft, and the 850 hPa temperature agreed with MEPS within about 1 K (Fig. 4); the landings met 15–24 kt at 1 000–2 000 ft against a 10 m wind of 7–9 kt (Fig. 5). If the model was right in the past hours, its next hours are more believable — and if it was off, the difference often persists for a while.

6 The values of one hour

Move the mouse over the chart (on the phone: tap or drag): a line marks the hour and the lines above the chart give all its values, in three parts:

1. the **time** and the **METAR** of the hour (with METAR): *Fri 6.11. 14 UTC · METAR 13:50: +3 / −2 °C · 300° 7 kt · Q1026 · vis ≥10 km · no ceiling*;
2. **Measured** — what the aircraft measured that hour: *850 hPa −3.7 °C (13–14 UTC, 100 values) · 850 hPa wind 000° 28 kt · freezing level 2 000 ft · landings 1 000 ft 324°/18 kt, 2 000 ft 344°/23 kt (26 landings)*;
3. each **model**, with its lead time (*MEPS +2 h*; *GFS +75 h*): the temperature (with the typical error in brackets), the dew point, the 850 hPa values, the freezing level, the wind in kt and m/s, the gust, the precipitation (*dry*; for GFS's 3-hour steps the mean), the pressure, the cloud cover, the cloud base or ceiling and the visibility.

7 How far can the forecast be trusted?

A forecast for twelve days ahead is drawn with the same sharp line as one for twelve hours. To show the difference, the server keeps every run's forecast for the airport and, when the time comes, checks it against the EFHK METAR. From the last **30 days** it computes for each **lead day** (day 1 = the first 24 hours of a run, day 2 the next . . .) how far off the forecast was.

The band. With Typical error on, a light band is drawn around the temperature and wind-speed lines: at that lead day, **8 in 10** of the checked hours were within it (the 80th percentile of the absolute error). It is narrow on the first days and widens further out (Fig. 6). With Both it follows MEPS over MEPS's hours and GFS after them; until enough MEPS runs have been checked, the MEPS hours use GFS's width of the same lead day. A lead day gets a band once at least 60 hours from 6 runs have been checked. The read-out adds it to the value: *+5.3 °C (±1.7)*, *wind 263° 8 kt (±3 kt)*.

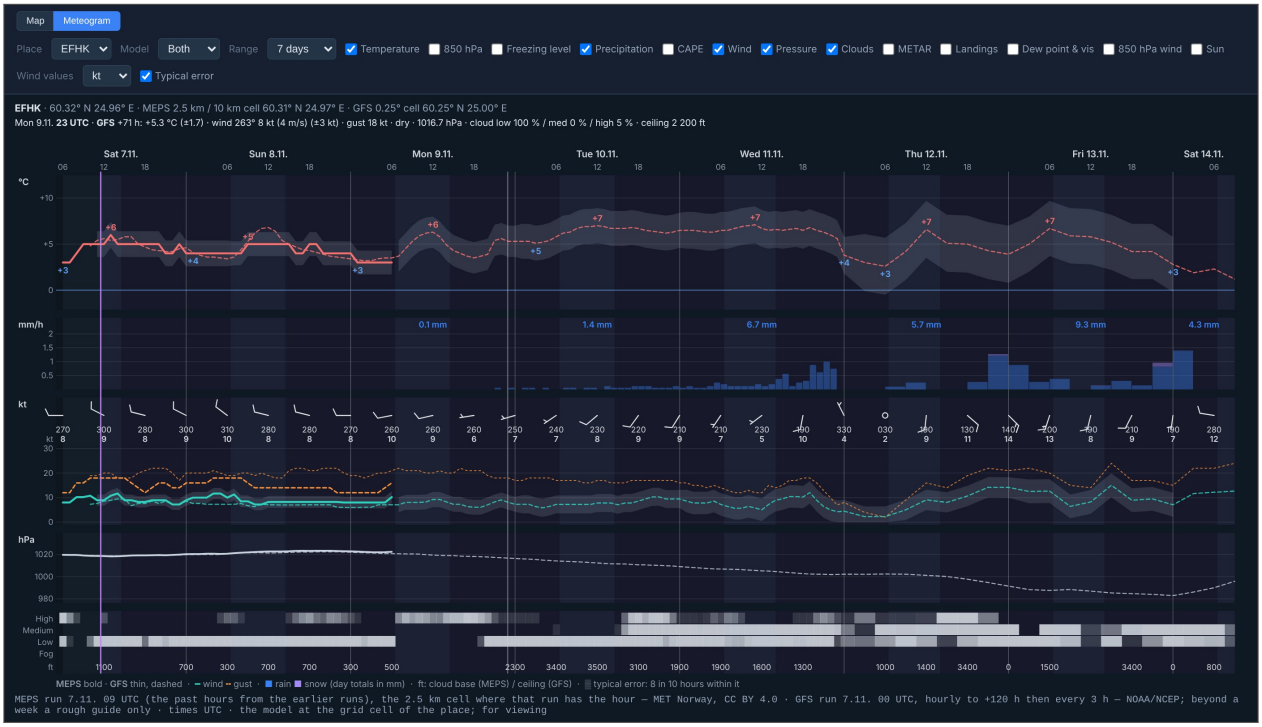
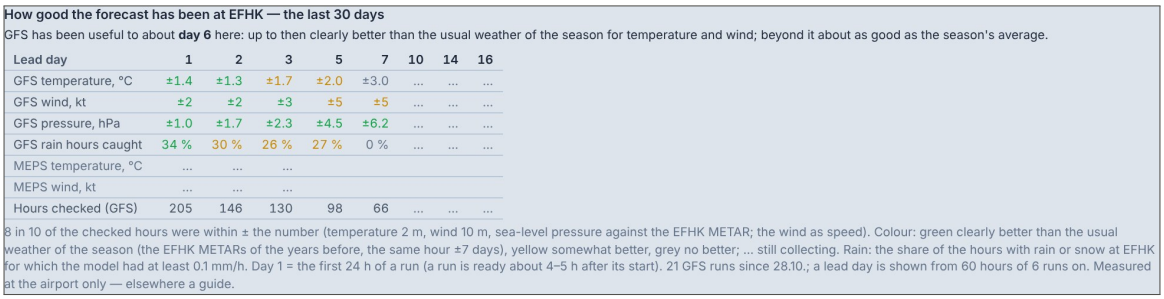


Figure 6. Both, 7 days, Typical error on and Wind values kt, dark theme; the live server's runs and forecast check of 7 November 2026, 11:32 UTC. The grey band around the temperature and wind lines widens from about ±1.4 °C on day 1 to ±3 °C on days 6–7; the wind band from about ±2 to ±5 kt. The mouse is on Monday 9 November 23 UTC: *GFS +71 h: +5.3 °C (±1.7)*.

The table under the chart (on the Weather page; Fig. 7) gives the numbers by lead day: the temperature, wind and pressure within which 8 in 10 hours fell, the share of the rain hours that the model caught (it had at least 0.1 mm/h when the METAR reported rain or snow), MEPS's first days and the number of hours checked. **The colours** compare the forecast with simply expecting *the usual weather of the season* — the EFHK METARs of the years before, at the same hour within ±7 days: **green** clearly better (the error at most half of the usual weather's), **yellow** somewhat better, **grey** no better; . . . still being collected. The sentence above the table sums it up: *GFS has been useful to about day N here*.



How good the forecast has been at EFHK — the last 30 days

GFS has been useful to about **day 6** here: up to then clearly better than the usual weather of the season for temperature and wind; beyond it about as good as the season's average.

Lead day	1	2	3	5	7	10	14	16
GFS temperature, °C	±1.4	±1.3	±1.7	±2.0	±3.0	…	…	…
GFS wind, kt	±2	±2	±3	±5	±5	…	…	…
GFS pressure, hPa	±1.0	±1.7	±2.3	±4.5	±6.2	…	…	…
GFS rain hours caught	34 %	30 %	26 %	27 %	0 %	…	…	…
MEPS temperature, °C	…	…	…					
MEPS wind, kt	…	…	…					
Hours checked (GFS)	205	146	130	98	66	…	…	…

8 in 10 of the checked hours were within ± the number (temperature 2 m, wind 10 m, sea-level pressure against the EFHK METAR; the wind as speed). Colour: green clearly better than the usual weather of the season (the EFHK METARs of the years before, the same hour ±7 days), yellow somewhat better, grey no better; … still collecting. Rain: the share of the hours with rain or snow at EFHK for which the model had at least 0.1 mm/h. Day 1 = the first 24 h of a run (a run is ready about 4–5 h after its start). 21 GFS runs since 28.10.; a lead day is shown from 60 hours of 6 runs on. Measured at the airport only — elsewhere a guide.

Figure 7. The forecast-check table of 7 November 2026, 11:32 UTC (21 GFS runs since 28 October). GFS's temperature was within ±1.4 °C in 8 of 10 hours on day 1 and ±2.0 °C on day 5 — clearly better than the season's usual weather on days 1–2, somewhat better to day 6 (hence *useful to about day 6*); the wind within ±2 kt on day 1 and ±5 kt on days 5–7; the pressure clearly better all week. Days 8 and on, and MEPS's first runs, were still being collected (…).

Reading it. The check is made at the airport only; for another place the band is a guide. 30 days is a short period: a calm, settled spell makes the forecast look better than a stormy one. On the phone the band is the same and one line under the chart sums up the table (Section 8).

8 On the phone

The phone's Meteogram view (Fig. 8) draws the same chart, smaller:

- **Chips** at the top: Place (EFHK, the receiver sites RX-01 …, Tapped), Model (MEPS, GFS, Both) and Range (2 d, **5 d**, 7 d, 10 d, 16 d; with MEPS only its own forecast).
- **Meteogram options** in the menu (the three-line button at the top left): a switch for each panel and option, and Wind values Off / kt / m/s as chips. Pressure is off by default on the phone.
- **Tap or drag** on the chart: the values of the hour appear above it.
- Instead of the table, **one line** under the chart: *GFS at EFHK, last 30 days: temperature ±1.4 °C on day 1, ±1.7 °C on day 3 … useful to about day 6*.

The view is refreshed every 10 minutes while it is open; each update is a few kB.

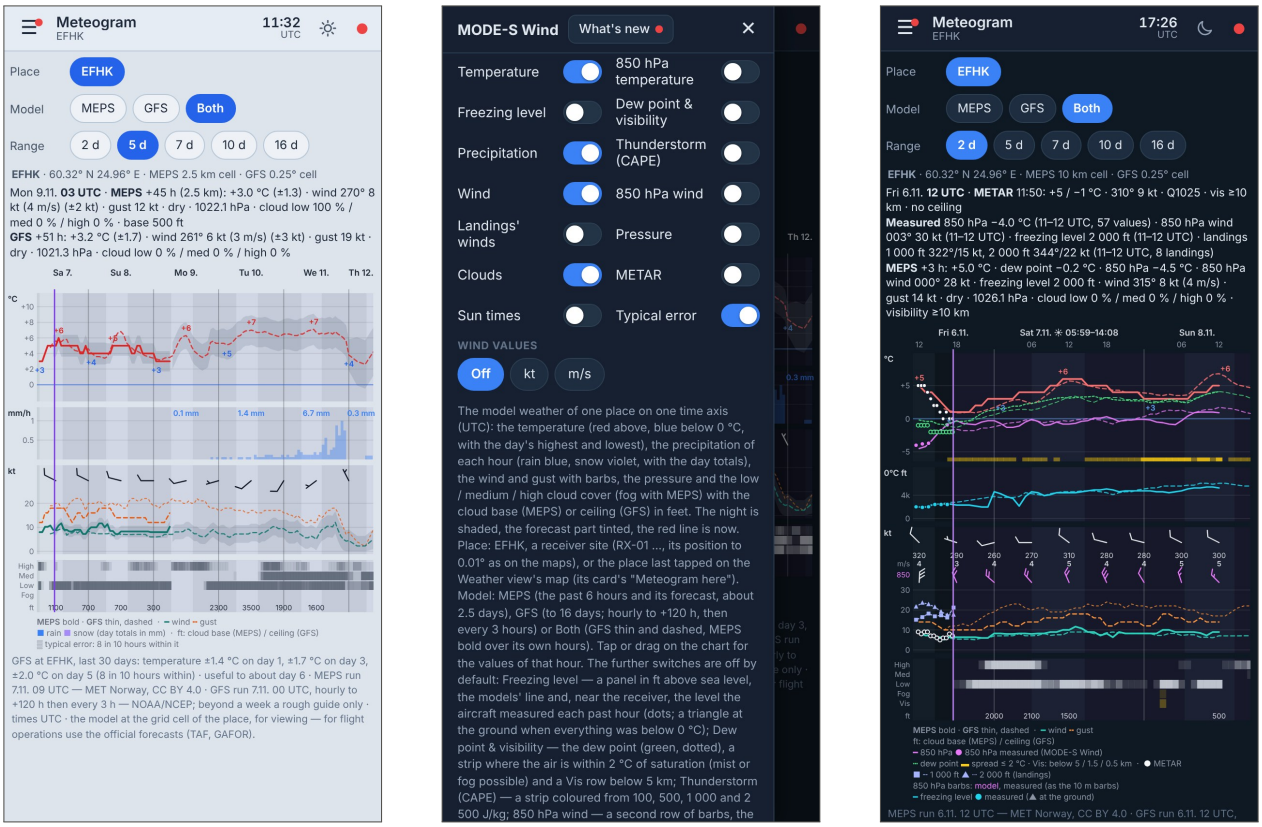


Figure 8. The phone. Left: Both, 5 d, light theme, with a tap on Monday 9 November 03 UTC (the live server's runs of 7 November 2026, 11:32 UTC): MEPS +45 h and GFS +51 h with their typical errors. Middle: the Meteogram options in the menu. Right: 6 November 2026 with the options on and Wind values m/s, dark theme: under the barbs the direction and the speed in m/s (*320 / 4*), with a tap on Friday 12 UTC.

9 In Finnish

For users with the **FIN** permission the meteograms — the Weather page's and the phone's — are in **Finnish**: the controls and their help texts, the weekdays (*ma ti ke* . . .), the panel labels, the legend, the read-out, the line under the chart and the table, with decimal commas (*+2,5 °C*). Times stay in UTC and the units are unchanged. The rest of the site stays in English. The permission is given by the maintenance user; a page already open shows the change after a reload. There is a Finnish version of this guide (*Meteogrammit*), and the meteogram's ? opens it for these users.

10 Reading a meteogram sensibly

- **A grid cell, not a point.** A coast, a lake, a hill or a town nearby can make the place's weather differ from its cell's — the wind and the temperature at night most of all.
- **Trust falls with the days.** MEPS is the better guide for the next two days; GFS's coarser grid smooths the showers and the wind. Beyond about a week GFS is a rough guide: look at the trend (colder, wetter, windier), not at the hour. The typical-error band shows how fast the trust falls.
- **Where the models part, the weather is uncertain.** With Both, a large difference between MEPS and GFS is information in itself.
- **Compare with what was measured.** The dots of the past hours show whether the model has got today right.
- **Precipitation is the hardest.** A shower can be right in time and a few kilometres off in place; the day total is more reliable than the hour.
- **UTC.** Add 2 hours for Finnish winter time, 3 in summer.
- **Not for flight planning.** Use the official TAF, GAFOR and the aviation weather briefing.

A Terms in English and Finnish

English	Finnish	English	Finnish
meteogram	meteogrammi	dew point	kastepiste
temperature	lämpötila	visibility	näkyvyys
freezing level	nollaraja	cloud base	pilven alaraja
precipitation	sade	ceiling	pilvikatto
rain / snow	vesi / lumi	high / medium / low cloud	ylä- / keski- / alapilvet
wind / gust	tuuli / puuska	fog	sumu
calm	tyyntä	measured	mitattu
wind barb	tuuliviiri	landings	laskeutumiset
wind values	tuulen arvot	typical error	tyypillinen virhe
pressure	paine	lead day	ennustepäivä
thunderstorm	ukkonen	range	jakso
dry	poutaa	model run	mallin ajo

Table 4. The meteogram's terms in English and in its Finnish version.

Data sources

- MEPS — MET Norway (MetCoOp), https://thredds.met.no, licence CC BY 4.0.
- GFS — NOAA / NCEP, https://nomads.ncep.noaa.gov, public domain.
- METAR EFHK — the aviation weather observations of Helsinki-Vantaa airport.
- The aircraft's temperatures and winds — MODE-S Wind's own receivers (Mode S EHS and ADS-B on 1 090 MHz).